

J MUKHESH

2303A51813

-- 1

```
SELECT emp_id, emp_name, dept_id, salary FROM employees e WHERE salary >
(SELECT AVG(salary) FROM employees WHERE dept_id = e.dept_id);
```

-- 2

```
SELECT emp_id, emp_name, dept_id, salary FROM employees e WHERE salary =
(SELECT MAX(salary) FROM employees WHERE dept_id = e.dept_id);
```

-- 3

```
SELECT MAX(salary) AS second_highest_salary FROM employees WHERE salary <
(SELECT MAX(salary) FROM employees);
```

-- 4

```
SELECT emp_name, salary FROM employees WHERE salary = (SELECT MAX(salary)
FROM employees WHERE salary < (SELECT MAX(salary) FROM employees));
```

-- 5

```
SELECT emp_id, emp_name, salary, manager_id FROM employees e WHERE
manager_id IS NOT NULL AND salary > (SELECT salary FROM employees WHERE
emp_id = e.manager_id);
```

-- 6

```
SELECT emp_id, emp_name, salary FROM employees e WHERE EXISTS (SELECT 1
FROM employees r WHERE r.manager_id = e.emp_id AND r.salary > e.salary);
```

-- 7

```
SELECT dept_id FROM employees GROUP BY dept_id HAVING AVG(salary) > (SELECT
AVG(salary) FROM employees);
```

-- 8

```
SELECT emp_id, emp_name, dept_id, salary FROM employees WHERE dept_id IN
(SELECT dept_id FROM employees GROUP BY dept_id HAVING AVG(salary) > (SELECT
AVG(salary) FROM employees));
```

-- 9

```
SELECT emp_id, emp_name, salary FROM employees WHERE salary > ALL (SELECT
salary FROM employees WHERE dept_id = (SELECT dept_id FROM departments WHERE
dept_name = 'HR'));
```

-- 10

```
SELECT emp_id, emp_name, salary FROM employees WHERE salary > ANY (SELECT
salary FROM employees WHERE dept_id = (SELECT dept_id FROM departments WHERE
dept_name = 'Research'));
```

-- 11

```
SELECT dept_id, dept_name FROM departments d WHERE NOT EXISTS (SELECT 1
FROM projects p WHERE p.dept_id = d.dept_id);
```

-- 12

```
SELECT emp_id, emp_name, dept_id FROM employees e WHERE (SELECT COUNT(*)
FROM projects p WHERE p.dept_id = e.dept_id) > 1;
```

-- 13

```
SELECT emp_id, emp_name, dept_id, salary FROM employees WHERE dept_id IN
(SELECT dept_id FROM projects WHERE budget = (SELECT MAX(budget) FROM
projects));
```

-- 14

```
SELECT emp_id, emp_name, salary FROM employees WHERE ABS(salary-(SELECT
AVG(salary) FROM employees)) = (SELECT MIN(ABS(salary-(SELECT AVG(salary) FROM
employees))) FROM employees);
```

-- 15

```
SELECT DISTINCT e.dept_id FROM employees e WHERE e.salary = (SELECT MAX(salary)
FROM employees WHERE dept_id = e.dept_id) AND e.salary > ALL (SELECT m.salary
FROM employees m WHERE m.dept_id = e.dept_id AND EXISTS (SELECT 1 FROM
employees r WHERE r.manager_id = m.emp_id));
```